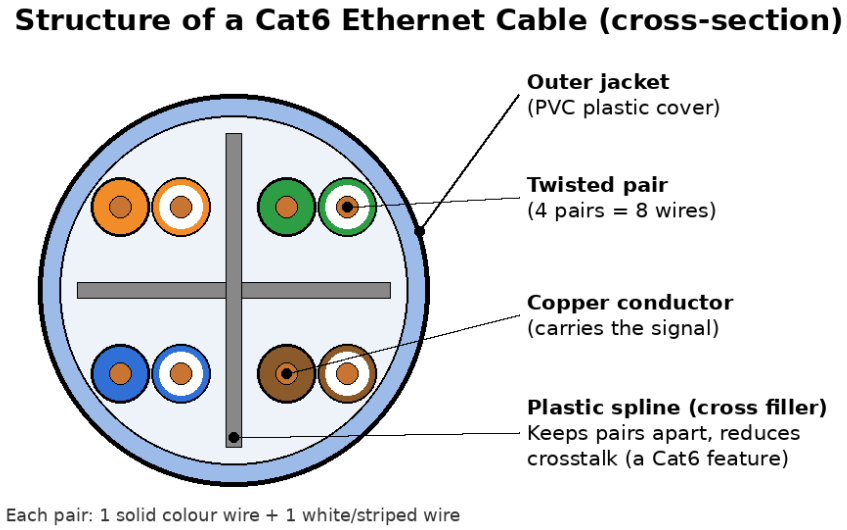
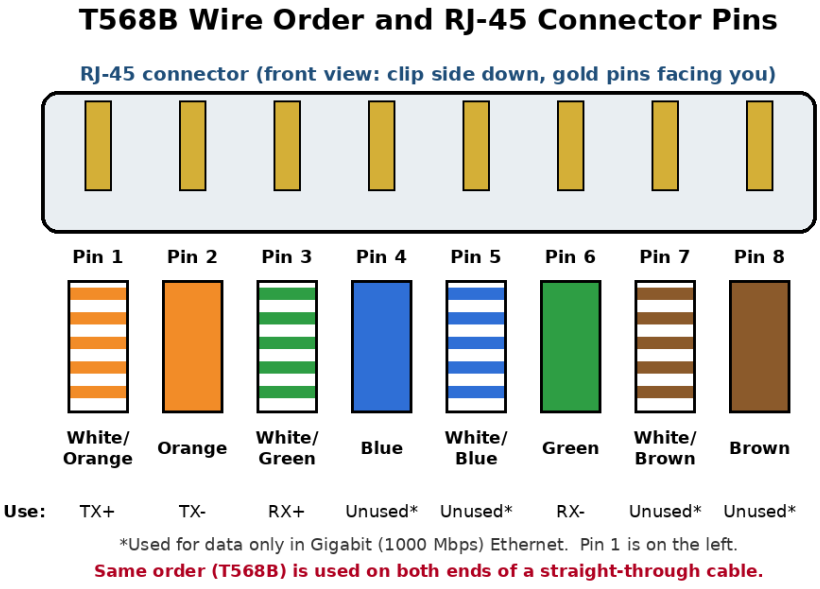
Network Cables and Connectors Assignment

Q1. Draw a labeled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labeling the RJ-45 connector pins.

Answer:





T568B color order (Pin 1 to 8): White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.

Q2. List three types of network cables (such as Cat5e, Cat6, and fiber optic) and for each, mention one real-world app or device (like a WiFi router, Zomato POS terminal, or IPL streaming server) where it would be used, explaining why.

Answer:

Cable Type	Where it is used (device / app)	Why
Cat5e (Ethernet)	WiFi router to a laptop or Zomato POS terminal (billing counter)	Cheap and supports up to 1 Gbps over 100 m, which is enough for billing and normal internet use.
Cat6 (Ethernet)	Switch to gaming consoles/PCs in a gaming zone, or office server connections	Gives stable gigabit speed and handles up to 10 Gbps on short runs (about 55 m). Less interference because of the internal spline.
Fiber optic	ISP to data center, or an IPL streaming server (Hotstar) connection	Uses light signals, so it is very fast, works over many kilometres and is not affected by electrical interference.

Q3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

Answer:

Tools needed: Cat6 cable, RJ-45 connectors, crimping tool, cable stripper, and a cable tester.

Steps:

1. Cut the cable to the length needed and remove about 2.5 cm (1 inch) of the outer jacket using the stripper, without cutting the wires inside.
2. Untwist the four pairs, straighten the wires and cut off the plastic spline.
3. Arrange the wires side by side in T568B order: White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.
4. Trim the wire ends evenly in a straight line, leaving about 1.2 cm (half inch) of wire out of the jacket.
5. Hold the RJ-45 connector with the clip facing down and push the wires in until they touch the front end. The outer jacket should go slightly inside the connector.
6. Place the connector in the crimping tool and squeeze firmly so the pins press into the wires.
7. Do the same on the other end with the same T568B order, then check the cable with a cable tester.

Two common mistakes to avoid:

- **Wrong color order:** Mixing up the wire order (or using different standards on both ends by mistake) makes the connection fail or slow.
- **Wires not pushed fully in / too much untwisting:** If the wires do not reach the end of the connector, or the pairs are untwisted too much, the connection becomes loose or noisy and may not work.

Q4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

Answer:

Connection	Cable	Connector	Why
Consoles / gaming PCs to the network switch	Cat6 (or Cat6a) Ethernet	RJ-45	Wired gigabit link, low delay (ping) and no WiFi drops during live matches.
Main switch to a second switch / router (long distance or backbone)	Fiber optic	LC connector (through SFP module)	Very high speed over long distance, no interference from many electronic devices.
Console / media player to each screen	High Speed HDMI (2.0 / 2.1)	HDMI	Carries 4K video and audio together in one cable.

Overall reasoning: Wired Ethernet is more stable and has lower delay than WiFi, which matters when many people are playing and streaming at once. Cat6 with RJ-45 is enough for consoles and gaming PCs, and fiber is used only where a long or very high-speed link is needed.

Q5. Find and share a photo (from your home, college, or online) of any network cable or connector in use (such as behind a WiFi router or at a café's POS counter), and describe what type it is and what devices it connects.

Answer:



Description: The photo shows a blue Cat5e/Cat6 Ethernet patch cable with clear plastic RJ-45 connectors. It is plugged into a LAN port at the back of the WiFi router and connects the router to a laptop (or desktop/smart TV). It is a twisted-pair copper cable that carries data at up to 1 Gbps.